

Forecasting USD/INR in the Indian Market with VMD, GDELT Agents, Monte Carlo, and TEPC

Repository summary of how the project moved from raw news signals toward bounded LLM overlays, market-memory systems, and topology-enabled predictions from chaos.

COURSE IN CHARGE
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POSTER FORMAT
A4 Portrait



RAW MARKET SIGNAL

0.010

Goldstein vs IMF3 is almost zero before feature engineering.

RECOVERED SIGNAL

-0.147

Theme-filtered Tone_Economy recovers useful structure.

V4 LLM RESULT

64.17%

Gemini V4 direction accuracy on 240 hybrid days.

TEPC RESULT

77.5%

Strict TEPC breakout accuracy; macro-F1 = 0.337.

1. PROBLEM

Why the Indian Market Is Hard to Predict

- USD/INR is a managed-float market shaped by RBI intervention, oil shocks, dollar cycles, and delayed news absorption.
- Raw sentiment is weak: Goldstein vs IMF3 = **0.010**; US average tone vs USD/INR = **0.164**.
- Delayed effects still appear: Goldstein peaks at **9 days** and tone at **12 days**.

Key insight: the market looks random in raw aggregates, but delayed and theme-specific structure appears after filtering.

2. VMD + STATS

VMD Reframes the Task from Price to Volatility

- Three-mode VMD targets **IMF3**, the high-frequency and news-sensitive part of USD/INR.
- Phase B uses four themes: **Economy, Conflict, Policy, Corporate**.
- Tone Economy** = **-0.147** with IMF3; the Phase-C danger-day classifier reaches **40% precision** and **52% recall**.

3. MONTE CARLO

Monte Carlo Supports a Noise-Dominant Market View

- Monte Carlo is most useful as a **distributional risk engine**, not a single-point predictor.
- Regime switching is the most balanced 30-day setup: median **90.370**, std **1.336**, 5th to 95th percentile **88.232 to 92.618**.

4. GDELT AGENTS

Multi-Agent LLMs Work Only as Bounded Overlays

V4 ACCURACY TABLE			
Model	Days	MAE	Dir.
V3 hybrid	240	0.2741%	61.25%
V4 hybrid	240	0.2695%	64.17%
V4 Jan live	21	0.3216%	57.14%

Debiasing mattered: symmetric prompting, randomized direction order, adaptive weights, and persona accountability improved V4 over V3.

5. TEPC

Most Recent Research: Topology-Enabled Predictions from Chaos

- TEPC treats USD/INR as one node in a macro-geopolitical network containing DXY, BRENT, GOLD, rates, sentiment, and Goldstein pressure.
- Rolling correlations define the graph; Laplacian summaries and coupled Lorenz chaos features become predictors.
- tepc_full** wins the strict raw-GDELT run: **77.5% breakout accuracy**, **0.337 macro-F1**, **0.002746 MAE return**. Live runs also show **chaos_only = 80%** breakout accuracy.

6. HYBRID TAKEAWAY

TEPC + LLMs for the Win Means Structured State + Controlled Reasoning

TEPC / LLM ACCURACY TABLE			
Model	Window	Acc.	Error
Memory-only state	100d	61.0% dir.	0.242 price
Gemini 2.5 personas	100d	58.0% dir.	0.272 price
TEPC full	80d	77.5% breakout	0.002 retur
TEPC persona	80d	78.75% breakout	0.002 retur

Final Message: the Indian FX market is not solved by a bigger black-box model. The strongest repo path is a layered pipeline: **VMD + thematic GDELT filtering + probabilistic checks + bounded multi-agent reasoning + TEPC topology/chaos features**.

Source: repo plots, run summaries, and research paper artifacts.

LLMs help most when they consume structured market memory, and TEPC adds nonlinear regime signal on top of that state.